KidsPortal

Maths Curriculum & Syllabus Guide

Grades 1 – 10 · A reference for parents

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About this guide

This document lists every topic your child covers in KidsPortal's Maths subject, grade by grade, so you can see exactly what is being taught and when. Each grade also includes a note on how that stage aligns with England's National Curriculum for Mathematics — the primary framework this subject is built on — with the roughly equivalent US grade band noted alongside for families comparing against Common Core Math standards.

These alignment notes describe the learning stage and skill focus each framework shares with that grade — not a word-for-word copy of either curriculum's official programme of study, since specific topics and exam boards vary by school.

Grade 1

Place value to 100, the four operations in their simplest form, fractions, measurement, money, time and shape.

1	Place Value to 100
2	Addition & Subtraction
3	Multiplication & Division
4	Fractions
5	Measurement
6	Money
7	Time
8	Shape
9	Position & Direction
10	Statistics
11	Counting & Number Bonds to 20
12	Doubling, Halving & Sharing
13	Comparing & Ordering Numbers
14	Odd & Even Numbers
15	Simple Addition & Subtraction Word Problems
16	Patterns & Sequences
17	Comparing Lengths & Heights
18	Counting Coins & Simple Money
19	Comparing Capacity (More/Less Full)
20	2D Shape Sorting
21	Telling Time to the Hour & Half Hour
22	Grouping & Counting in Tens

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 1 (Year 1) National Curriculum for Mathematics — counting, number bonds and simple addition/subtraction — broadly equivalent to US Kindergarten early-numeracy expectations.

Grade 2

Place value to 1,000, formal addition and subtraction, the 3, 4 and 8 times tables, fractions of amounts, and simple data handling.

1	Place Value to 1,000
2	Formal Addition & Subtraction
3	Times Tables: 3, 4 & 8
4	Tenths & Equivalent Fractions
5	Perimeter
6	Money: Problem Solving
7	Time
8	Angles & Shape Properties
9	Bar Charts & Tables
10	Reasoning & Problem Solving
11	Multiplication & Division Facts: 2, 5 & 10
12	Fractions of Shapes & Amounts
13	Comparing Mass & Capacity
14	3 & 4 Times Tables
15	2D Shape Properties: Sides & Vertices
16	Telling Time to 5 Minutes
17	Division as Sharing Equally
18	Introducing Halves & Quarters
19	Money: Adding Coins and Notes
20	Measuring Length in cm and m
21	Weighing Objects in Grams and Kilograms
22	Simple Tally Charts and Data

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 1 (Year 2) National Curriculum for Mathematics, including the statutory 2, 5 and 10 times tables — broadly equivalent to US Grade 1 Common Core Math standards.

Grade 3

Place value to 10,000, negative numbers, times tables to 12×12, decimals, area and perimeter, and multi-step word problems.

1	Place Value to 10,000
2	Negative Numbers
3	Times Tables to 12×12
4	Decimals
5	Area & Perimeter
6	Money & Decimals
7	Converting Units of Time
8	Symmetry & Coordinates
9	Discrete & Continuous Data
10	Multi-Step Word Problems
11	Formal Column Addition & Subtraction
12	Adding & Subtracting Fractions
13	Telling Time: Digital & 24-Hour Clock
14	Multiplying 2-Digit by 1-Digit Numbers
15	Perimeter of Simple Shapes
16	Bar Charts & Pictograms
17	Equivalent Fractions
18	Measuring in Litres & Millilitres
19	Telling Time: the 24-Hour Clock
20	Finding Area by Counting Squares
21	Multi-Step Word Problems
22	Introducing Roman Numerals

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 2 (Year 3) National Curriculum for Mathematics and its full times-tables-to-12×12 expectation — broadly equivalent to US Grade 2 Common Core Math standards.

Grade 4

Place value to a million, formal written multiplication and division, fractions and decimals, perimeter/area/volume, and line graphs.

1	Place Value to 1,000,000
2	Roman Numerals
3	Formal Written Multiplication & Division
4	Adding & Subtracting Fractions
5	Decimals & Percentages
6	Perimeter, Area & Volume
7	Angles in Shapes
8	Position & Transformation
9	Line Graphs
10	Problem Solving & Reasoning
11	Multiplying & Dividing by 10, 100 & 1,000
12	Rounding & Estimating
13	Properties of Triangles & Quadrilaterals
14	Introducing Decimals: Tenths & Hundredths
15	Area of Rectangles
16	Negative Numbers on a Number Line
17	Adding & Subtracting Fractions with the Same Denominator
18	Line Symmetry
19	Rounding Decimals
20	Decimals in Money and Measurement
21	Plotting Coordinates (First Quadrant)
22	Translating Shapes

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 2 (Year 4) National Curriculum for Mathematics, where formal written methods for all four operations are introduced — broadly equivalent to US Grade 3 Common Core Math standards.

Grade 5

Place value to 10 million, the four operations with larger numbers, ratio and proportion, introductory algebra, and compound shapes.

1	Place Value to 10,000,000
2	The Four Operations
3	Fractions, Decimals & Percentages
4	Ratio & Proportion
5	Introduction to Algebra
6	Metric & Imperial Conversion
7	Compound Shapes
8	Classifying Shapes
9	Pie Charts & the Mean
10	Reasoning Papers
11	Prime Numbers, Factors & Multiples
12	Multiplying & Dividing Fractions
13	Volume of Cubes & Cuboids
14	Introducing Ratio
15	Angles in a Straight Line & Around a Point
16	Mean, Median, Mode & Range
17	Percentages of Amounts
18	Plotting Coordinates in All Four Quadrants
19	Converting Between Fractions, Decimals & Percentages
20	Volume of Cubes and Cuboids
21	Finding the Mean (Average)
22	Simple Formulae and Substitution

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 2 (Year 5) National Curriculum for Mathematics, including the first introduction to algebraic notation — broadly equivalent to US Grade 4 Common Core Math standards.

Grade 6

Algebraic expressions, ratio and proportion, angle and shape properties, probability basics, and solving linear equations.

1	Number
2	Algebraic Expressions
3	Fractions, Decimals & Percentages
4	Ratio & Proportion
5	Angles & Shape Properties
6	Averages & Range
7	Probability Basics
8	Sequences
9	Solving Linear Equations
10	Area, Perimeter & Volume
11	Negative Numbers & Order of Operations (BIDMAS)
12	Percentages: Increase & Decrease
13	Constructions & Loci
14	Simplifying Algebraic Expressions
15	Circumference & Area of a Circle
16	Pie Charts
17	Solving One-Step & Two-Step Equations
18	Volume of Cuboids
19	Nets of 3D Shapes
20	Ratio and Proportion Problems
21	Combined Probability Events
22	Negative Number Operations

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 2 (Year 6) National Curriculum for Mathematics, the final year of primary maths before secondary school — broadly equivalent to US Grade 5 Common Core Math standards.

Grade 7

Indices and standard form, Pythagoras' theorem, scatter graphs and correlation, transformations, and equations and inequalities.

1	Indices & Standard Form
2	Expanding & Factorising
3	Ratio, Proportion & Rates of Change
4	Pythagoras' Theorem
5	Scatter Graphs & Correlation
6	Combined Probability Events
7	Linear Graphs & Coordinates
8	Transformations
9	Equations & Inequalities
10	Compound Measures
11	Direct & Inverse Proportion
12	Congruence & Similarity
13	Solving Quadratic Equations by Factorising
14	Plotting & Interpreting Linear Graphs (y = mx + c)
15	Probability Basics
16	Standard Form
17	Finding the nth Term of Linear Sequences
18	Solving Linear Inequalities
19	Compound Measures: Speed, Density & Pressure
20	Factorising Simple Expressions
21	Scatter Graphs and Correlation
22	Congruence and Similarity

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 3 (Year 7) National Curriculum for Mathematics, where secondary pupils meet standard form and formal proof-style reasoning — broadly equivalent to US Grade 6 Common Core Math standards.

Grade 8

Surds, quadratic expressions, simultaneous equations, introductory trigonometry, circle theorems and vectors — GCSE foundations.

1	Surds & Standard Form
2	Quadratic Expressions
3	Simultaneous Equations
4	Pythagoras & Introduction to Trigonometry
5	Sampling & Data Analysis
6	Probability Trees & Venn Diagrams
7	Quadratic & Real-Life Graphs
8	Circle Theorems
9	Vectors
10	GCSE Foundations Review
11	Trigonometric Graphs: Sine, Cosine & Tangent
12	Algebraic Fractions
13	Histograms & Frequency Density
14	Solving Simultaneous Linear Equations
15	Introduction to Vectors
16	Pythagoras' Theorem in 3D
17	Expanding Two Brackets
18	Introduction to Trigonometry (SOHCAHTOA)
19	Direct & Inverse Proportion
20	Solving Simultaneous Equations Graphically
21	Circle Area and Circumference Problems
22	Standard Form Calculations

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 3 (Year 8) National Curriculum for Mathematics; many schools begin GCSE-specification topics like circle theorems and vectors here, as this unit does — broadly equivalent to US Grade 7 Common Core Math standards.

Grade 9

Standard form, quadratics, circle theorems, trigonometry beyond right angles, sampling and data, and functions and graphs.

1	Number: Standard Form, Surds & Indices
2	Algebra: Expanding, Factorising & Quadratics
3	Ratio, Proportion & Rates of Change
4	Geometry: Circle Theorems
5	Geometry: Trigonometry Beyond Right Angles
6	Statistics: Sampling & Data Representation
7	Probability: Combined Events & Tree Diagrams
8	Algebra: Simultaneous Equations
9	Vectors
10	Functions & Graphs
11	The Sine Rule & Cosine Rule
12	Completing the Square
13	Compound Interest & Growth/Decay
14	Function Notation
15	Transformations of Graphs
16	Surds
17	Solving Quadratic Equations by Factorising
18	Circle Theorems Introduction
19	Trigonometric Graphs
20	Simplifying Algebraic Fractions
21	Introduction to the Sine and Cosine Rules
22	Introduction to Histograms

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Still within Key Stage 3 (Year 9) nationally, though schools commonly start GCSE Maths content here to spread the specification over three years, as this unit does — broadly equivalent to US Grade 8 Common Core Math standards.

Grade 10

Algebraic proof, 3D geometry, cumulative frequency and box plots, conditional probability, and an introduction to differentiation and integration.

1	Algebra: Algebraic Fractions & Proof
2	Number: Bounds & Error Intervals
3	Geometry: 3D Shapes, Volume & Surface Area
4	Geometry: Similarity & Congruence
5	Statistics: Cumulative Frequency & Box Plots
6	Probability: Conditional Probability
7	Algebra: Iterative Methods & Graph Transformations
8	Ratio: Direct & Inverse Proportion
9	Transformations of Functions
10	Exam Revision: Problem-Solving & Mixed Practice
11	Vectors: Geometric Proof
12	Advanced Circle Theorems
13	Simultaneous Equations: One Linear, One Quadratic
14	Introduction to Differentiation
15	Cumulative Frequency & Box Plots
16	Algebraic Proof
17	Differentiation of Polynomials (Further Practice)
18	Integration Basics
19	Vector Geometry Problems
20	Differentiation Applications: Finding Maximum and Minimum Points
21	Composite and Inverse Functions
22	Introduction to Iterative Methods

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 4 (GCSE Year 1) (Year 10) — GCSE Maths content including calculus foundations (differentiation and integration), usually reserved for higher-tier papers — broadly equivalent to US Grade 9 (Freshman) Common Core Math standards for that band.